2.5 Acid–Base Titrations

Problems Worksheet

1. A primary standard is the starting point for a titration.
   a. Explain the role of a primary standard in volumetric analysis.

   b. Why is NaOH not suitable for use as a primary standard?

   c. Explain why a high molar mass is preferable for a primary standard.

   d. Why should a primary standard be stored in a desiccator before it is weighed?

   e. Explain what is meant by a secondary standard and give an example of how it is produced.

2. Determine the mass of solid required to make the following primary standard solutions:
   a. 0.100 mol.L^{-1} oxalic acid in a 500 mL volumetric flask. Oxalic acid available for use as a primary standards is a dihydrate with the formula $H_2C_2O_{4.2}H_2O$.

   b. 0.0500 mol.L^{-1} sodium carbonate (Na_2CO_3) in a 250 mL volumetric flask.

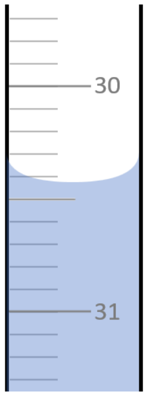
3. Explain the purpose of the following equipment used in titrations:
   a. Volumetric flask

   b. Pipette

   c. Burette

4. Mason is going to determine the concentration of an approximately 0.1 mol.L^{-1} HCl solution by titrating it against his primary standard solution of Na_2CO_3. He is putting the HCl in the burette and the Na_2CO_3 in the conical flask.
    a. When making the primary standard, Mason struggles to weigh out the exact mass he needs. The mass of Na_2CO_3 he adds to his 250 mL volumetric flask is 2.625 g. What is the concentration of his primary standard?
    b. What should he use to rinse the pipette and the burette?
    c. If he rinses the conical flask with the Na_2CO_3 solution, how will this affect the titration?
    d. If he decided to put the acid in the conical flask and the base in the burette, how would this affect the results on the titration?

5. When performing a titration, a reading needs to be taken from the burette at the start and at the end.
   a. Why is the burette read from the bottom of the meniscus?



   b. What is the reading on this burette?

   c. What is the uncertainty on this burette reading?

6. Explain the difference between the end point and the equivalence point in a titration.

7. For each of the following titrations, state the approximate pH at the equivalence point and state which indicator(s) would be most suitable.

	2	3	4	5	6	7	8	9	10
methyl orange									
bromothymol blue									
phenolphthalein									

   a. HCl and Na_2CO_3

   b. KOH and HCl

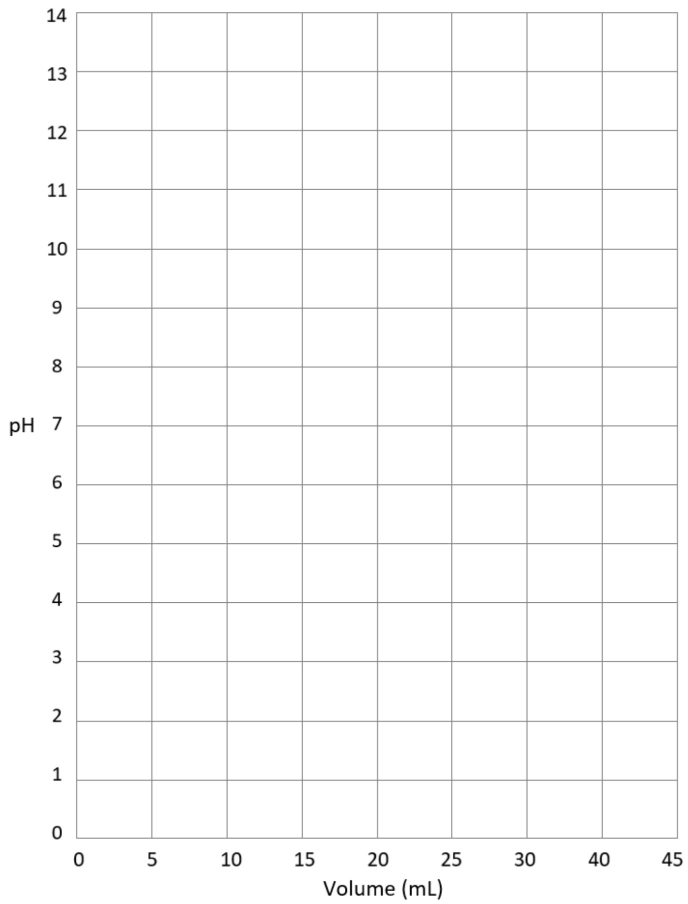
   c. NaOH and CH_3COOH

   d. $H_2C_2O_4$ and NaOH

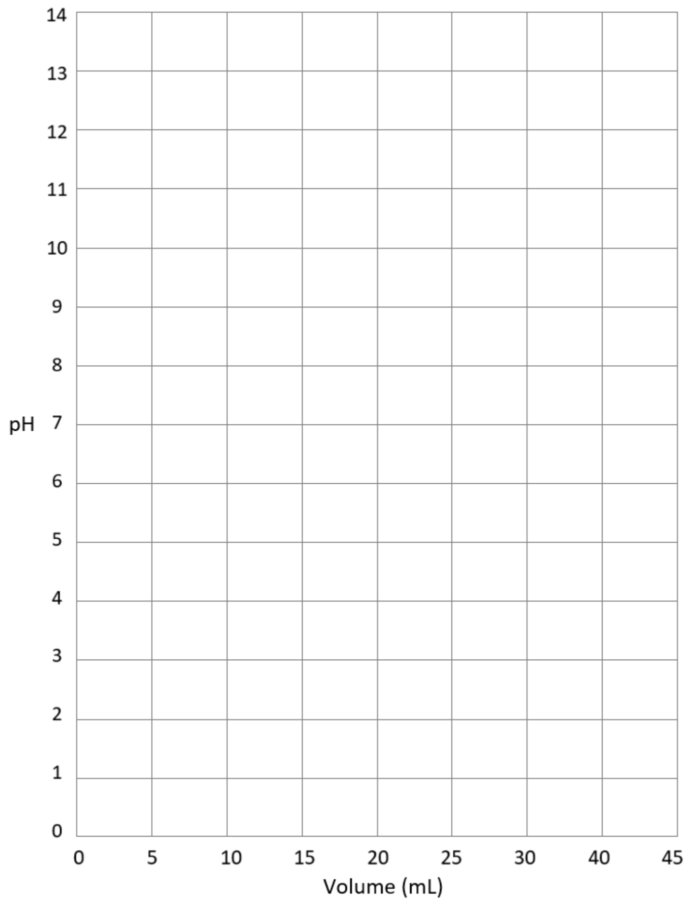
8. Performing a titration between a weak acid (such as CH_3COOH) and a weak base (such as NH_3) is difficult.
   a. Why is this more challenging than a titration where a strong acid or base is involved?

   b. How can this titration be performed if it is necessary?

9. Sketch a titration curve for the following titrations. Assume that the equivalence point occurs at 20 mL :
   a. KOH and HCl with the acid in the burette.



b. NaOH and CH_3COOH with the base in the burette.



c. $H_2C_2O_4$ and NaOH with the acid in the burette.